

COURSE TITLE : MICROCONTROLLER AND INTERFACING
COURSE CODE : 4214
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER: 56
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Introduction to 8051 microcontroller	14
2	Assembly language programming	19
3	Interfacing ICs	10
4	Interfacing of 8051	13
TOTAL		56

COURSE OUT COME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To know the difference between microprocessor and micro controller
	2	To comprehend the architecture of microprocessor 8085
	3	To comprehend the architecture of microcontroller 8051
	4	To understand the pin details of 8051
2	1	To comprehend the instruction set of 8051
	2	To understand timing subroutines
	3	To comprehend the look-up tables
	4	To understand the serial data communication
	5	To know the programming 8051
3	1	To understand architecture of 8255
	2	To comprehend architecture of 8251
	3	To understand architecture of 8259
	4	To understand architecture of 8279
4	1	To know the applications of 8051 in display devices
	2	To know the applications in A/D & D/A converters
	3	To understand the applications of 8051 in instrumentation
	4	To understand the applications in dc motors
	5	To understand the applications in stepper motors

SPECIFIC OUTCOME

MODULE I INTRODUCTION TO 8051 MICROCONTROLLER

1.1.0 To know the difference between microprocessor and micro controller.

- 1.1.1 To compare microprocessors and microcontrollers.
- 1.1.2 To list the various microcontroller manufacturers.

1.2.0 To comprehend the architecture of microprocessor 8085.

- 1.2.1 To describe the features of 8085.
- 1.2.2 To explain the internal block diagram of 8085.
- 1.2.3 To explain registers of 8085.

1.3.0 To comprehend the architecture of microcontroller 8051.

- 1.3.1 To describe the features of 8051.
- 1.3.2 To explain the programming model of microcontroller 8051.
- 1.3.3 To illustrate register structure of 8051.
- 1.3.4 To explain special function registers of 8051.
- 1.3.5 To describe internal memory map of 8051
- 1.3.6 To illustrate the ports of 8051
- 1.3.7 To explain timer mode of operation
- 1.3.8 To explain interrupts in 8051
- 1.3.9 To describe the stack in 8051
- 1.3.10 To describe the power saving modes of operation

1.4.0 To understand the pin details of 8051

- 1.4.1 To list the I/O pins
- 1.4.2 To explain the dual functions of I/O pins
- 1.4.3 To describe other various pin details

MODULE II ASSEMBLY LANGUAGE PROGRAMMING

2.1.0 To comprehend the instruction set of 8051

- 2.1.1 To explain the addressing modes
- 2.1.2 To classify the instructions
- 2.1.3 To explain the arithmetic, logic instructions
- 2.1.4 To illustrate the Rotate instructions
- 2.1.5 To describe Jump and call instructions

2.2.0 To understand timing -subroutines

- 2.2.1 To describe the working of subroutines
- 2.2.2 To explain the working of timing subroutine
- 2.2.3 To list the applications of timing subroutine

2.3.0 To comprehend the look-up tables

- 2.3.1 To explain the applications of look-up tables
- 2.3.2 To state the need of look-up tables in the program
- 2.3.3 To illustrate the creation of Look-up table

2.4.0 To understand the serial data communication

- 2.4.1 To explain the modes of serial data communication
- 2.4.2 To describe the serial data transmission
- 2.4.3 To explain the serial data reception

2.5.0 To know the programming of 8051

- 2.5.1 To familiarise the assembly language programming
- 2.5.2 To explain the programmes like largest in array, sorting, Code conversion, scale conversion, unit conversion

MODULE III INTERFACING ICS

3.1.0 To understand architecture of 8255

- 3.1.1 To describe the architecture of 8255
- 3.1.2 To list the pins of 8255
- 3.1.3 To explain the control register of 8255
- 3.1.4 To explain the modes of 8255
- 3.1.5 To illustrate the ports of 8255

3.2.0 To comprehend architecture of 8251

- 3.2.1 To describe the architecture of 8251
- 3.2.2 To list the pins of 8251
- 3.2.3 To Explain the control register of 8251

3.3.0 To understand architecture of 8259

- 3.3.1 To describe the architecture of 8259
- 3.3.2 To list the pins of 8259
- 3.3.3 To explain the status register of 8259

3.4.0 To understand architecture of 8279

- 3.4.1 To describe the architecture of 8279
- 3.4.2 To list the pins of 8279
- 3.4.3 To explain the display section of 8279
- 3.4.4 To illustrate the keyboard section of 8279
- 3.4.5 To describe the key board debounce and control
- 3.4.6 To explain the Scan section of 8279

MODULE IV INTERFACING OF 8051

4.1.0 To know the applications of 8051 in display devices

- 4.1.1 To explain the pins of LCD display IC LM051
- 4.1.2 To explain the interfacing of LCD using time delay
- 4.1.3 To describe the commands in LCD
- 4.1.4 To illustrate the 7 segment display
- 4.1.5 To list the types of 7 segment display
- 4.1.6 To explain the interfacing of 7 segment display

4.2.0 To know the applications in A/D & D/A converters

- 4.2.1 To explain the pins of A/D converter IC ADC0804
- 4.2.2 To illustrate the interfacing of A/D converter
- 4.2.3 To describe the pins of D/A converter IC
- 4.2.4 To explain the interfacing of D/A converter

4.3.0 To understand the applications of 8051 in instrumentation.

- 4.3.1 To describe the interfacing of temperature IC LM 35.

4.4.0 To understand the applications in dc motors

- 4.4.1 To describe the interfacing of DC motor

4.5.0 To understand the applications in Stepper motors

- 4.5.1 To Define step angle
- 4.5.2 To Classification of stepper motor
- 4.5.3 To Describe the control circuits
- 4.5.4 To Describe the interfacing of stepper motors

CONTENT DETAILS

MODULE I

Comparison of microprocessor and microcontroller - features of 8085- internal architecture of 8085 registers Of 8085- features of 8051 - port the features of 8051 - the programming model of microcontroller 8051- register structure of 8051- special function register of 8051- internal-memory map of 8051 - the ports of 8051 - timer mode of operation- interrupts in 8051- the stack in 8051- power saving modes of operation in 8051- I/O pins-the dual functions of I/O pins- other pin details

MODULE II

Instruction set of 8051 - addressing modes – arithmetic ,logic instructions - Rotate instructions - Jump and call instructions - timing –subroutines - applications of timing subroutine - look-up tables & application - serial data communication and various modes – transmission and reception of serial data - programming of 8051

MODULE III

Architecture of 8255 – pins - control register- modes and ports - architecture of 8251 – pins – control register - architecture of 8259 – pins – control register - architecture of 8279 - pins – display section of 8279 - keyboard section of 8279 - scan section of 8279

MODULE IV

Interfacing with 8051 - LCD system - 4x4 keyboard - ADC and DAC - stepper motor - dc motor - temperature control system.

REFERENCES

1. Subratha Ghoshal , 8051 microcontroller internals, instructions, programming and interfacing , Pearson
2. Muhammad Ali Mazidi and Janice Gillispie Mazidi , The 8051 Microcontroller and Embedded Systems Using Assembly and C - Second Edition ,Pearson Education
3. Kenneth J Ayala - Thomson ,The 8051 Microcontroller , Third Edition
- 4.Ramesh S Gaonkar , Microprocessor Programming and applications ,New Age publications